

Replacing Physical Records & Storage with a Centralised, Segmented Server Network for Every School Department

■ TABLE OF CONTENTS

1.	Why Schools Need a Server — The Problem Statement
2.	What Is a School Server? Architecture Overview
3.	Server OS Comparison (Ubuntu · Windows Server · TrueNAS)
4.	Available Server Options & Cost Breakdown
5.	Network Topology — Departments & Segmentation (VLANs)
6.	Step-by-Step: Install Ubuntu Server 24.04 LTS
7.	CLI Configuration — IP, DNS, DHCP, Samba, FTP, NFS
8.	Samba File Share — Department Folders
9.	Active Directory / LDAP — User Accounts per Department
10.	Backup & Maintenance Schedule
11.	Plug-and-Use Guide for Staff (Non-Technical)
12.	Troubleshooting Checklist
13.	Screenshot Checklist — What to Capture
14.	Demonstration Scenarios
15.	Social-Media Quotes for Reddit / LinkedIn / GitHub
16.	Quick-Reference Card

1 WHY SCHOOLS NEED A SERVER

The Problem With Paper and USB Drives

Most schools still rely on paper registers, USB drives, individual desktop hard drives, and filing cabinets to manage student records, staff documents, timetables, exam results, fee receipts, library catalogues, and departmental data. This creates serious operational and security risks.

Problem	Impact
Lost/damaged paper records	Permanent data loss — no recovery
USB drives passed between staff	Virus spread, accidental deletion
No central access	Admin must physically move to each PC
No backup system	A single hard-drive failure = all records gone
No audit trail	Cannot tell who changed or deleted a file
Physical space consumed by cabinets	Wasted room, pest damage, fire risk
Slow retrieval of student info	Minutes to hours finding one record
No concurrent access	Two staff cannot edit the same file
No role-based access	Student could walk up and open staff files

■ A centralised school server solves ALL of the above problems while costing less than a new photocopier — and it lasts 10+ years.

2 SCHOOL SERVER ARCHITECTURE

High-Level Network Diagram (Text)

[illegible]

Department Shares Layout on the Server

VLAN	Department	Share Path	Who Can Access
VLAN 10	Administration	/shares/admin	Principal, Deputy, Registrar
VLAN 20	Academic / Teachers	/shares/academic	All teaching staff
VLAN 30	Students	/shares/students	Read-only class resources
VLAN 40	Library	/shares/library	Librarian, Students (read)
VLAN 50	Finance	/shares/finance	Bursar, Accounts only
VLAN 60	IT / Backups	/shares/it	IT Admin only

3 SERVER OS COMPARISON

Which Operating System Is Best for a School Server?

Feature	Ubuntu Server 24.04 LTS	Windows Server 2022	TrueNAS SCALE
Cost	FREE	\$500–\$6,000 licence	FREE
Learning curve	Medium (CLI-heavy)	Low (GUI)	Low (Web UI)
File sharing	Samba (SMB/CIFS)	Native SMB	SMB + NFS + iSCSI
Active Dir.	OpenLDAP / Samba AD	Built-in AD DS	Limited
Backup	rsync + cron	Windows Backup	ZFS snapshots
Security	UFW firewall + AppArmor	Windows Defender	ZFS + RBAC
Hardware req.	1 GB RAM min (4 rec)	4–16 GB RAM	8 GB RAM (ZFS)
Best for	Most schools ■	Existing MS infra	Storage-only NAS
Community	Huge / free support	Paid Microsoft support	Growing

■ **RECOMMENDATION:** Use Ubuntu Server 24.04 LTS as the primary server. It is free, secure, lightweight, and has 5 years of Long-Term Support until 2029. Pair it with TrueNAS SCALE as a dedicated NAS/backup appliance if budget allows.

4 AVAILABLE SERVERS & COST BREAKDOWN

Recommended Hardware Options (2025 Pricing)

Option	Hardware Spec	Est. Cost (USD)	Best For
Budget	Used Dell PowerEdge R320 / HP ProLiant DL160 Xeon E5, 16 GB RAM, 2×1 TB HDD	\$150–\$400	Schools under 200 students
Mid-Range	New Dell PowerEdge T150 Tower Xeon E-2300, 32 GB RAM, 4×2 TB HDD RAID	\$800–\$1,400	Schools 200–800 students
Enterprise	Dell PowerEdge R450 / HPE ProLiant DL380 2× Xeon, 64 GB RAM, 6×4 TB SAS RAID	\$2,500–\$5,000	Large schools / multi-campus
Cloud-Hybrid	Raspberry Pi 4 (8 GB) local + AWS/GCP cloud Offsite backup + remote access	\$80 Pi + ~\$20/mo cloud	Very small / rural schools

Additional Cost Items

Item	Approx Cost (USD)	Notes
Managed Network Switch (24-port)	\$80–\$300	Required for VLAN segmentation
UPS (Uninterruptible Power Supply)	\$120–\$350	Protects server from power cuts
Cat6 Ethernet cabling per room	\$30–\$80/room	Needed for wired connections
Wireless Access Points (per AP)	\$60–\$200	For departments needing Wi-Fi
External backup HDD (4 TB)	\$80–\$120	Weekly offline backup drive
Ubuntu Server 24.04 LTS OS	\$0	Completely free, open source
Samba / OpenLDAP / DNS software	\$0	All free, open source
IT setup labour (once-off)	\$200–\$500	Or self-configure with this guide
Annual maintenance	\$100–\$300/yr	UPS battery, dust, updates

5 NETWORK TOPOLOGY & VLAN SEGMENTATION

Why Segment the Network?

Without VLANs, every device on the school network can see every other device — meaning a student's laptop could potentially access the Finance department's files. VLANs (Virtual Local Area Networks) create logical walls between departments. Each department is isolated; traffic only passes between VLANs when the firewall/router explicitly allows it.

VLAN Configuration on a Managed Switch (CLI Example)

MANAGED SWITCH – VLAN SETUP (Cisco IOS / Compatible Switches)

```
# Connect to switch via SSH or console cable
ssh admin@192.168.1.1

# Enter configuration mode (Cisco-style IOS)
enable
configure terminal

# Create VLANs
vlan 10
name ADMIN
vlan 20
name ACADEMIC
vlan 30
name STUDENTS
vlan 40
name LIBRARY
vlan 50
name FINANCE
vlan 60
name IT_SERVERS

# Assign switch port Gi0/1 to VLAN 10 (Admin room PC)
interface GigabitEthernet0/1
switchport mode access
switchport access vlan 10
no shutdown
```

```
# Trunk port to server (carries all VLANs)
interface GigabitEthernet0/24
switchport mode trunk
switchport trunk allowed vlan 10,20,30,40,50,60

# Save config
end
write memory
```

6 INSTALLING UBUNTU SERVER 24.04 LTS

Step 1 — Download & Create Boot USB

Download Ubuntu Server 24.04 LTS from ubuntu.com/download/server. Use Balena Etcher or Rufus to write the ISO to a USB drive (8 GB+).

Step 2 — Boot from USB and Follow Installer

1. Insert USB into server, reboot, press F11/F12 to select boot device
2. Choose: 'Install Ubuntu Server' → Language: English → Keyboard: your layout
3. Network: select the NIC, set static IP (see below)
4. Storage: use entire disk → set up LVM (recommended) → confirm
5. Profile: Full name, server name (e.g. school-server), username (e.g. itadmin), password
6. SSH: ☒ Install OpenSSH server (important — enables remote CLI access)
7. Snaps: skip optional snaps — we'll install manually
8. Reboot when prompted, remove USB

Step 3 — First Login and System Update

UBUNTU SERVER — FIRST LOGIN

Login as itadmin at the console or SSH from another machine:

```
ssh itadmin@192.168.60.10
```

Update all packages

```
sudo apt update && sudo apt upgrade -y
```

Set the hostname

```
sudo hostnamectl set-hostname school-server
```

Verify

```
hostname
```

```
# Output: school-server
```

7 CLI CONFIGURATION — IP, DNS, DHCP

7.1 Static IP with Netplan

UBUNTU — NETPLAN STATIC IP

```
# Edit the netplan config
sudo nano /etc/netplan/00-installer-config.yaml

# Paste this content (adjust interface name with: ip link show)
network:
version: 2
renderer: networkd
ethernets:
ens3: # Change to your NIC name
dhcp4: no
addresses:
- 192.168.60.10/24 # Server static IP (IT VLAN)
nameservers:
addresses: [8.8.8.8, 1.1.1.1]
routes:
- to: default
via: 192.168.60.1 # Your router/gateway IP

# Apply the config
sudo netplan apply

# Verify IP address
ip addr show ens3
```

7.2 Install & Configure DNSMasq (Local DNS + DHCP)

UBUNTU — DNSMASQ DNS + DHCP

```
# Install dnsmasq
sudo apt install dnsmasq -y

# Backup default config
sudo cp /etc/dnsmasq.conf /etc/dnsmasq.conf.bak

# Write new config
```

```
sudo nano /etc/dnsmasq.conf

# Paste the following:
# ████████████████████████████████████████████████████████████

interface=ens3
bind-interfaces

# DHCP ranges per VLAN
dhcp-range=192.168.10.100,192.168.10.200,24h # Admin
dhcp-range=192.168.20.100,192.168.20.200,24h # Academic
dhcp-range=192.168.30.100,192.168.30.200,24h # Students
dhcp-range=192.168.40.100,192.168.40.200,24h # Library
dhcp-range=192.168.50.100,192.168.50.200,24h # Finance

# Local DNS records
address=/school-server.local/192.168.60.10
address=/files.school.local/192.168.60.10
# ████████████████████████████████████████████████████████████

# Enable and start
sudo systemctl enable dnsmasq
sudo systemctl restart dnsmasq

# Check status
sudo systemctl status dnsmasq
```

8 SAMBA FILE SHARE — DEPARTMENT FOLDERS

What is Samba?

Samba allows Linux servers to share folders with Windows, macOS, and Linux computers using the standard SMB/CIFS protocol — the same protocol Windows uses natively. Every PC in the school can map a network drive (e.g. Z:\) that points to the server.

SAMBA — INSTALL & CREATE SHARES

```
# Install Samba
sudo apt install samba samba-common-bin -y

# Create department directories
sudo mkdir -p /shares/{admin,academic,students,library,finance,it}

# Set permissions (admin group owns /shares/admin, etc.)
sudo groupadd school-admin
sudo groupadd school-academic
sudo groupadd school-students
sudo groupadd school-library
sudo groupadd school-finance
sudo groupadd school-it

sudo chown -R root:school-admin /shares/admin
sudo chown -R root:school-academic /shares/academic
sudo chown -R root:school-students /shares/students
sudo chown -R root:school-library /shares/library
sudo chown -R root:school-finance /shares/finance
sudo chown -R root:school-it /shares/it

sudo chmod -R 2770 /shares/admin
sudo chmod -R 2770 /shares/academic
sudo chmod -R 2775 /shares/students # read for all students
sudo chmod -R 2770 /shares/library
sudo chmod -R 2770 /shares/finance
sudo chmod -R 2770 /shares/it
```

SAMBA — /etc/samba/smb.conf

```
# Backup default smb.conf
```

comment = Student Resources (Read Only)

```
# .....  
.....
```

SAMBA – ADD USERS

9 OPENLDAP — CENTRALISED USER ACCOUNTS

OpenLDAP provides a centralised directory of all user accounts. Instead of creating separate logins on every PC, staff and students log in once and the server authenticates them. Think of it as Active Directory for Linux.

OPENLDAP — INSTALL & CONFIGURE

```
# Install OpenLDAP
sudo apt install slapd ldap-utils -y

# During install: set admin password when prompted
# Reconfigure if needed:
sudo dpkg-reconfigure slapd
# → DNS domain: school.local
# → Organisation name: My School
# → Admin password: (strong password)
# → Database: MDB
# → Remove database on purge: No
# → Move old database: Yes

# Verify LDAP is running
sudo systemctl status slapd

# Test: query the directory
ldapsearch -x -H ldap://localhost -b dc=school,dc=local

# Install LDAP Account Manager (web GUI – optional but recommended)
sudo apt install ldap-account-manager -y
# Access at: http://192.168.60.10/lam
```

10 BACKUP & MAINTENANCE

10.1 Automated Backup with rsync + cron

BACKUP – rsync AUTOMATED SCRIPT

[illegible]

10.2 Maintenance Schedule

Frequency	Task	Command / Action
Daily (auto)	Backup all shares	cron rsync script
Weekly	Check disk space	df -h /shares
Weekly	Review Samba logs	sudo tail -100 /var/log/samba/smb.log
Weekly	Check failed logins	sudo grep 'authentication failure' /var/log/auth.log
Monthly	Apply security patches	sudo apt update && sudo apt upgrade -y
Monthly	Test backup restore	Copy 1 file from /backup to /tmp and verify
Monthly	Check UPS battery	Physical inspection / UPS software
Termly	Review user accounts	sudo pdbedit -L wc -l
Termly	Delete old accounts	sudo userdel -r former.student
Yearly	Full hardware check	Dust fans, check HDD health: sudo smartctl -a /dev/sda

11 PLUG-AND-USE GUIDE FOR STAFF

Once the IT administrator has set up the server, staff do not need any technical knowledge. Follow these steps on any Windows PC in the school.

Connecting a Windows PC to the School File Server

Step	Action	What to Type / Click
1	Open File Explorer	Press Win + E
2	Go to Network location	Click 'This PC' → 'Map network drive'
3	Choose a drive letter	Select Z: (or any unused letter)
4	Enter server address	\\school-server\Academic (or Admin, Students, etc.)
5	Tick 'Reconnect at sign-in'	Ensures drive maps automatically on login
6	Enter credentials	Username: mrs.okonkwo Password: (your Samba password)
7	Done!	Z: drive now appears in File Explorer — use like a normal folder

■ *TIP FOR IT ADMIN: Use a Group Policy Object (GPO) or a login script to map network drives automatically for all PCs. Then staff never need to do this manually — the drive just appears when they log in.*

Connecting a Linux / Ubuntu Desktop

LINUX CLIENT – MOUNT SAMBA SHARE

```
# Install cifs utils
sudo apt install cifs-utils -y

# Create mount point
sudo mkdir -p /mnt/school/academic

# Mount the share
sudo mount -t cifs //192.168.60.10/Academic /mnt/school/academic \
-o username=mrs.okonkwo,password=YourPassword,uid=1000,gid=1000

# To auto-mount on boot – add to /etc/fstab:
//192.168.60.10/Academic /mnt/school/academic cifs \
username=mrs.okonkwo,password=YourPassword,uid=1000,gid=1000,_netdev 0 0
```

12 TROUBLESHOOTING CHECKLIST

Problem	Likely Cause	Fix Command
Cannot ping server	Wrong IP / firewall	ping 192.168.60.10 sudo ufw allow from 192.168.0.0/16
Cannot access share (Access Denied)	User not in Samba group	sudo usermod -aG school-academic username sudo smbpasswd -e username
Share not visible on Windows	NetBIOS / nmbd not running	sudo systemctl restart nmbd sudo systemctl status nmbd
Samba service down	Config error	testparm sudo journalctl -u smbd -n 50
Disk full	Shares over capacity	df -h /shares du -sh /shares/*
Slow network speeds	Switch duplex mismatch	ethtool ens3 sudo ethtool -s ens3 speed 1000 duplex full
DHCP not assigning IPs	dnsmasq misconfigured	sudo systemctl status dnsmasq sudo journalctl -u dnsmasq -n 30
SSH connection refused	SSH not installed / port blocked	sudo apt install openssh-server sudo ufw allow 22
Wrong time on server	NTP not synced (affects logs/AD)	sudo apt install systemd-timesyncd timedatectl status
Hard drive failure warning	SMART error on HDD	sudo apt install smartmontools sudo smartctl -H /dev/sda

13 SCREENSHOT CHECKLIST

Take these screenshots to document your project for Reddit / LinkedIn / GitHub:

#	What to Screenshot	How to Get There	Why It's Important
1	Ubuntu Server boot login prompt	Boot the server — the login screen with hostname	Proves server is live
2	ip addr show — showing static IP	After: ip addr show ens3	Shows network config
3	Samba shares listed in Windows File Explorer	On Windows PC: open \\school-server	Visual proof of shares
4	Mapped Z: drive with department folder	After mapping Z: drive in Windows	Shows plug-and-use
5	Creating a file in share from Windows	Use test.txt from Windows, cat /shares/academic/testcrossplatformer	Cross-platform demo
6	df -h output showing share disk usage	Run: df -h /shares	Shows disk monitoring
7	Samba user list: pdbedit -L	Run: sudo pdbedit -L	Shows user management
8	testparm output (no errors)	Run: testparm	Config validation
9	Cron job for backup: crontab -l	Run: sudo crontab -l	Automated backup proof
10	rsync backup directory listing	Run: ls /backup/	Backup exists
11	Firewall rules: ufw status numbered	Run: sudo ufw status numbered	Security config
12	Switch VLAN config output	On managed switch: show vlan brief	Network segmentation
13	LDAP Account Manager web UI	Open http://192.168.60.10/lam	User directory GUI
14	Access Denied when Finance folder opened by student account	Log opened by student account \\school-server\Finance	Security demo
15	Backup log file tail	Run: tail -20 /var/log/school-backup.log	Logging proof

14 DEMONSTRATION SCENARIOS

Scenario A — Student Tries to Access Finance Files (Security Demo)

■ *Goal: Show that VLAN + Samba permissions prevent unauthorised access.*

SCENARIO A – SECURITY DEMO (ACCESS DENIED)

```
# On a student PC (VLAN 30), try to access Finance share:
net use F: \\school-server\Finance /user:student.john MyPassword

# Expected output:
System error 5 has occurred.
Access is denied.

# Also try from Linux:
smbclient //192.168.60.10/Finance -U student.john
# Output: NT_STATUS_ACCESS_DENIED

# Screenshot this – it's your security proof.
```

Scenario B — Teacher Saves Exam Paper, Another Teacher Opens It

■ *Goal: Show real-time file sharing between two staff on the same share.*

SCENARIO B – REAL-TIME FILE SHARING

```
# Teacher 1 (mrs.okonkwo) on PC-1 – creates a file:
echo "Grade 10 Maths Exam - Term 2" > Z:\Exams\maths_t2_2025.txt

# Teacher 2 (mr.adeyemi) on PC-2 – reads the same file immediately:
type Z:\Exams\maths_t2_2025.txt
# Output: Grade 10 Maths Exam - Term 2

# On the server you can see both users connected:
sudo smbstatus

# Screenshot: smbstatus showing two active connections
```

Scenario C — IT Admin Checks Disk Space and Backs Up

SCENARIO C – BACKUP DEMONSTRATION

```
# Check disk space
df -h /shares
```

```
# Filesystem Size Used Avail Use% Mounted on
# /dev/sda1 2.0T 340G 1.7T 18% /shares

# Run manual backup
sudo /usr/local/bin/school-backup.sh

# Confirm backup created
ls -lh /backup/
# drwxr-xr-x 2025-05-26 2025-05-26/

# Check backup log
tail -5 /var/log/school-backup.log
# sent 4,732,156 bytes received 82 bytes 9,464,476.00 bytes/sec
# Backup completed: 2025-05-26
```

Scenario D — New Staff Account Created in Under 2 Minutes

SCENARIO D — NEW STAFF ONBOARDING IN 2 MINUTES

```
# IT Admin creates new teacher account
sudo useradd -m -s /bin/bash mr.ibrahim
sudo usermod -aG school-academic mr.ibrahim
sudo smbpasswd -a mr.ibrahim
# New SMB password: .....

# Verify account exists
sudo pdbedit -L | grep ibrahim
# mr.ibrahim:1003:Mr Ibrahim

# New teacher immediately maps drive from their PC — DONE!
# No reinstall, no IT visit needed.
```

15 SOCIAL MEDIA QUOTES & CAPTIONS

Reddit Posts

[r/sysadmin](#) · [r/selfhosted](#) · [r/linux4noobs](#)

I set up a Linux Samba server for a local school using a \$300 used Dell PowerEdge. Replaced 20 years of paper records and USB drives. Every department now has its own network share, VLANed and permission-locked. Cost: \$300 hardware + \$0 software. Full guide with CLI commands inside.
[OC]

[r/homelab](#) · [r/techsupport](#)

PSA: Schools don't need expensive proprietary software for file management. Ubuntu Server + Samba + OpenLDAP = a full Active-Directory-style file server, completely free. Took 4 hours to set up. Here's the full step-by-step.

[r/networking](#)

VLANed a school network today. Students on VLAN 30 literally cannot even see the Finance folder — it doesn't appear in the browse list. This is why network segmentation matters even in small environments.

LinkedIn Posts

■ *Transforming how schools manage data — one server at a time.*

Last month I deployed a centralised Ubuntu Server for a secondary school that had been storing student records in filing cabinets and on USB sticks for over a decade.

The result? Every department has its own secure network share. Files are backed up automatically every night. New staff accounts take 2 minutes to

create.

The entire server hardware cost less than a new photocopier.

If your school is still using physical storage, let's talk.

**#EdTech #SysAdmin #Linux #SchoolIT #DigitalTransformation #Samba
#Ubuntu**

■ *You don't need a big budget to modernise school IT infrastructure.*

Here's what a \$400 server running Ubuntu + Samba + OpenLDAP can do:

- *Centralise all school records*
- *Give each department its own secure folder*
- *Block students from accessing staff or finance data*
- *Auto-backup every night*
- *Add a new user account in 2 minutes*

*I've written a complete step-by-step guide with CLI commands.
Link in comments.*

#Linux #OpenSource #SchoolIT #NetworkAdmin #TechForGood

GitHub README / Project Description

School Server Infrastructure Project

> A complete, free, open-source guide to deploying a centralised Ubuntu Server for schools — replacing paper records and USB drives with a secure, VLAN-segmented Samba file server, complete with CLI commands, cost breakdown, OS comparison, demo scenarios, and a maintenance schedule.

****Stack:** Ubuntu Server 24.04 LTS · Samba · OpenLDAP · dnsmasq · rsync · UFW**

*****Cost:** \$0 software · \$150–\$1,400 hardware***

*****Audience:** IT admins, school tech teams, sysadmin students, r/selfhosted community***

16 QUICK REFERENCE CARD

Essential Commands for Daily IT Admin

Command	Purpose
sudo systemctl status smbd	Check if Samba is running
sudo systemctl restart smbd nmbd	Restart Samba
sudo pdbedit -L	List all Samba users
sudo smbpasswd -a username	Add new Samba user
sudo smbpasswd -x username	Delete a Samba user
sudo usermod -aG school-academic user	Add user to Academic group
smbstatus	Show who is connected right now
testparm	Test smb.conf for errors
df -h /shares	Check disk space on shares
du -sh /shares/*	Size of each department share
sudo ufw status numbered	Check firewall rules
sudo ufw allow from 192.168.20.0/24	Allow VLAN 20 through firewall
sudo smartctl -H /dev/sda	Check hard drive health
sudo journalctl -u smbd -n 50	Last 50 Samba log lines
sudo apt update && sudo apt upgrade -y	Update all software packages
sudo crontab -l	View scheduled backup jobs
sudo /usr/local/bin/school-backup.sh	Run manual backup now
tail -20 /var/log/school-backup.log	View recent backup log
sudo netplan apply	Apply network config changes
ip addr show	Show all IP addresses
ping 192.168.60.10	Test server reachability

"A school that cannot find a student's record in under 10 seconds does not have an information system — it has an information disaster."

— Deploy your school server. Today. For free.